

Andrew Barchuk

SOFTWARE ENGINEER

Berlin, Germany

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"The direction we're heading is more important than where we're standing now."

Summary

A software engineer with more than 8 years of professional experience focusing primarily on building backend services. The particular areas of interest include *distributed systems, concurrency, scalability and performance, monitoring and observability*. Passionate about constantly improving the way we build systems, deepening my understanding of technology, optimizing software delivery and developer productivity, always looking for opportunities to improve.

Skills and technologies

programming	Java, Rust, Scala, Go, C++, Python, JavaScript, Bash
data storage	Kafka, ZooKeeper, Cassandra, Spanner, PostgreSQL, MongoDB
software design	software architecture and design patterns, microservices, CQRS; algorithm analysis and implementation, concurrency
web	HTTP and design of RESTful APIs, WebSockets, understanding of underlying networking stack (TCP, UDP, DNS)
performance	profiling and optimization: heap analysis, CPU sampling, JVM GC, benchmarking, load testing
testing	TDD practitioner, JUnit, Mockito, property-based testing
DevOps	Kubernetes, Docker; AWS and GCP clouds; monitoring (Prometheus, Grafana, distributed tracing); Unix toolset, Linux administration, ZFS
security	securing APIs, OAuth, symmetrical/asymmetrical cryptography, GPG
languages	English, German (A2), Ukrainian, Russian

Experience

Zalando

Berlin, Germany

PRINCIPAL SOFTWARE ENGINEER, RISK AND ABUSE MANAGEMENT

Oct. 2022 - Present

- Lead a multi-year effort to build a near-real-time fraud detection platform. Define a foundation for MLOps risk infrastructure.
- Design Zalando invitation-only purchase model. Co-design design of and lead the data migration to a event-sourced order storage. Contribute to the design of the the central ML & AI platform. Facilitate adoption of Apache Flink as the company-standard stream processing framework.
- Responsible for technology vision, target architecture, and techology stack of a three team department.

SENIOR/LEAD SOFTWARE ENGINEER, CHECKOUT BACKEND

Oct. 2020 - Oct. 2022

- Maintain and evolve business critical purchase systems, including zero-downtime migration from Cassandra to AWS DynamoDB. Responsible for performance, reliability and tech debt management. Help to establish the newly formed team.

Revolut

Berlin, Germany

SENIOR SOFTWARE ENGINEER, OPEN BANKING

Jun. 2020 - Sep. 2020 (4 months)

- Data synchronization reliability, transaction enrichment; Java, PostgreSQL, Spring

Google

Zurich, Switzerland

SOFTWARE ENGINEER, YOUTUBE CHANNEL MEMBERSHIPS

Jul. 2018 - Mar. 2020 (1 year, 9 months)

- Subscription based, user supported YouTube alternative monetization (C++, Spanner, PubSub, CQRS, Stubby). Projects: iOS payments, optimizing channel members retrieval latency (100x for large channels), pricing levels.
- 20% project: Renaming of symbols in object files (ELF, Mach-O, PE/COFF) in Rust for Mundane cryptographic library.

Tink

Stockholm, Sweden

SOFTWARE ENGINEER, CORE PLATFORM

Oct. 2016 - Jun. 2018 (1 year, 8 months)

- Core transaction processing platform development (Java and Go, Cassandra, MySQL, Kafka, ZooKeeper, Elasticsearch, AWS, Prometheus, Grafana, Kibana, SaltStack provisioning, Terraform-defined declarative infrastructure). Projects: up/down scaling of transaction processing, driving movement towards microservices, continues delivery pipeline, improving transaction processing performance, advocating for testing, working with infrastructure closely.

Playtika

Kyiv, Ukraine

JAVA DEVELOPER, INFRASTRUCTURE SERVICES

Mar. 2016 - Oct. 2016 (8 months)

- Service for messages delivery via WebSockets to the end users (Spring Boot, Apache Kafka, Couchbase).
- Facebook ad audiences management service: in memory processing of large amounts of data and integration with Marketing API.

Yandex

Kyiv, Ukraine

SOFTWARE ENGINEER, MOVIE PLATFORM

Apr. 2014 - Mar. 2016 (2 years)

- Backend development for integrating kinopoisk.ru (movie catalogue website) with Yandex services. Working on high load (order of 10K QPS) system based on Java 8, Spring Framework, Jetty, MongoDB and MySQL. Designing (public and internal) REST APIs. Working on authentication and authorization. Load testing, performance optimization, JVM profiling. Facilitating team TDD adoption. Participating in software development best practices promotion and implementation as a team. Mentoring of new project members.

DIO-Soft

Kyiv, Ukraine

JAVA DEVELOPER, STATE STREET BANK

Nov. 2013 - Mar. 2014 (5 months)

- Development of a backend for financial data monitoring service (Restlet Framework, Spring DI, JAXB, Sybase, SQL, JDBC, Java concurrency, TDD, JUnit, Mockito, integration testing).

NetCracker

Kyiv, Ukraine

JUNIOR SOFTWARE ENGINEER, SYSTEM INTEGRATION

Mar. 2013 - Oct. 2013 (8 months)

- Client integration projects (Java, SOAP web services, PL/SQL, JSP, JavaScript).

Open source contributions

Mundane, Rust cryptographic library (<https://github.com/google/mundane>)

BUILD SYSTEM IMPROVEMENTS, ELF/MACH-O, COFF MANIPULATION

2019

- Reimplement listing of exported symbols in object files in Rust, remove dependency on Go from build system.
- Replace two-step build process with in-place symbol renaming (ongoing).

goblin, Rust object file parsing library (<http://github.com/m4b/goblin>)

MACH-O, AND LIBRARY ARCHIVE PARSER IMPROVEMENTS

2019

- Prevent parser crashes on certain Mach-O files.
- Implement parsing of Windows-specific library archives.
- Implement detection of static library archive type.

TestNG, Java testing framework (<http://github.com/cbeust/testng>)

IMPROVE CONSISTENCY OF ARRAY COMPARISON

2015

- Make sure arrays are compared in the same way in all assertions.

Talks

Lessons Learned From Building the Fashion Store Checkout Data Platform

Berlin, Germany

ZALANDO INTERNAL

2025

Distributed coordination: ZooKeeper and Chubby

Zurich, Switzerland

GOOGLE INTERNAL

2018

Distributed Locking with MongoDB

Kyiv, Ukraine

YANDEX JAVA PARTY

2015

Education

Igor Sikorsky Kyiv Polytechnic Institute

Kyiv, Ukraine

BACHELOR OF SOFTWARE ENGINEERING

2010-2015